Exploratory Data Analysis (EDA) Summary Report

Geldium Finance – Credit Card Delinquency Risk | Prepared by Helga Yuliani Putri Aritonang, AI Transformation Consultant, Tata iQ

1. Introduction

Geldium Finance has seen a rise in credit card delinquency and wants to move from manual, history-based case handling to a data-driven way of identifying at-risk customers. Before any predictive model is built, this report assesses whether the customer dataset is complete, consistent and informative enough to support delinquency prediction. Its goals are to (1) describe the structure and quality of the data, (2) document missing values and how they should be treated, and (3) identify early risk indicators and data gaps that the Analytics team must address before modeling.

2. Dataset Overview

Key dataset attributes:

- **Number of records:** 500 customers and 19 columns; no duplicate rows or duplicate Customer_IDs.
- **Target variable:** Delinquent_Account (1 = delinquent). 80 customers (16.0%) are delinquent and 420 (84.0%) are not, so the classes are imbalanced.
- **Key variables:** Credit_Score, Credit_Utilization, Missed_Payments (last 12 months), Debt_to_Income_Ratio, Income, Loan_Balance, Account_Tenure, Employment_Status, Credit_Card_Type, Location, Age, and a six-month payment history (Month_1 to Month_6: On-time, Late or Missed).
- **Data types:** 9 numerical (Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Account_Tenure, Delinquent_Account) and 10 categorical (Customer_ID, Employment_Status, Credit_Card_Type, Location, Month_1 to Month_6).

Inconsistencies and anomalies found in the initial review:

- **Inconsistent category labels:** Employment_Status uses "Employed" (82), "employed" (77) and "EMP" (81) for the same group; once merged, 240 customers are employed.
- **Unit mismatch with the data guide:** Credit_Utilization and Debt_to_Income_Ratio are stored as decimals (0–1), not percentages (0–100%). Four customers have utilization above 1.0 (maximum 1.026), meaning they are over their credit limit.
- **Values clustered at a floor:** 9 customers have utilization of exactly 0.05 and 14 have a debt-to-income ratio of exactly 0.10, which suggests capping or default values rather than real observations.
- **Unexpected categories:** Credit_Card_Type includes "Student" (112) and "Business" (108), which are not in the data guide (Standard, Gold, Platinum).
- **Logical conflicts:** For 141 customers (28.2%), Missed_Payments over 12 months is lower than the number of "Missed" months in the last six months alone, which is impossible. In addition, 46 customers have a credit score of 750 or more alongside 4

or more missed payments, 14 "retired" customers are under 30, and 81 customers (16.2%) have an account tenure longer than their adult years (Age − 18).

- **Fields that do not agree with each other:** Debt_to_Income_Ratio is barely related to Loan_Balance ÷ Income (Spearman ρ = 0.08), and credit score is unrelated to missed payments (ρ = −0.01). In real credit data these move together.

Initial data quality summary. The dataset is largely complete: only three fields have gaps, and none exceed 8% of records. Its main weakness is consistency rather than completeness. Category labels need standardizing, units differ from the documentation, and several fields contradict each other in ways that would not occur in genuine credit records. Most importantly, the payment-behavior fields that should drive delinquency prediction do not agree with each other, so the definition and source of each field should be confirmed with Geldium before modeling.

3. Missing Data Analysis

Key missing data findings:

- **Variables with missing values:** Income 39 (7.8%), Loan_Balance 29 (5.8%) and Credit_Score 2 (0.4%). Four customers are missing both Income and Loan_Balance. The six-month payment history is complete.
- **Missingness mechanism:** Customers with missing Income do not differ from others in delinquency rate (12.8% vs 16.3%, Fisher p = 0.82) or in age, credit score, utilization, missed payments or debt-to-income (all p > 0.2). This is consistent with data missing completely at random (MCAR), although missing-not-at-random cannot be ruled out from the observed data alone. Missing Loan_Balance is linked to a lower debt-to-income ratio (0.25 vs 0.30, p = 0.006), so it is missing at random (MAR) with respect to that variable, and these customers show a higher, though not statistically significant, delinquency rate (24.1% vs 15.5%, p = 0.29).
- **Missing data treatment:** Imputation with missing-value flags. Deletion was rejected because dropping incomplete rows would remove 64 customers (12.8%), 11 of them delinquent, from a dataset with only 80 positive cases. Synthetic data was not used for imputation because the gaps are small and simple, transparent methods are enough.

Missing data issue	Handling method	Justification
Income – 39 missing (7.8%)	Median imputation ($107,658) plus an Income_Missing flag	Distribution is near-symmetric (mean $108,380 vs median $107,658), so the median is robust and transparent; the flag lets the model detect if missingness itself signals risk.
Loan_Balance – 29 missing (5.8%)	Median imputation ($45,776) plus a Loan_Missing flag	Regression imputation was tested but rejected: loan balance is unrelated to income or debt-to-income (|ρ| < 0.06), so it would add complexity without accuracy; the flag preserves the higher delinquency rate in this group.
Credit_Score – 2 missing (0.4%)	Median imputation (586)	With only two records affected, the choice has negligible impact on the distribution, and keeping the rows avoids losing data.

4. Key Findings and Risk Indicators

Key findings:

- **Correlations observed between key variables:** No variable has a statistically meaningful relationship with delinquency. Every numerical feature has a correlation with Delinquent_Account below 0.05 in absolute value (all $p > 0.3$), and no categorical feature differs significantly across groups (employment $p = 0.55$, card type $p = 0.35$, location $p = 0.66$). As a baseline check, a logistic regression and a random forest using all features achieved a cross-validated AUC of only 0.44 and 0.42, no better than random guessing (0.50).
- **Unexpected anomalies:** Customers with 0–1 missed payments have a slightly higher delinquency rate (18.3%) than those with 4–6 missed payments (16.0%), and customers with credit scores above 650 are delinquent slightly more often (17.7%) than those with scores of 500 or below (16.2%). Both run against basic credit logic and, together with the field conflicts in Section 2, suggest that Delinquent_Account may be defined or recorded differently from the behavioral fields. This needs investigation by the Analytics team before any model is trusted.

High-risk indicators to carry into modeling (based on credit-risk practice; current evidence in this dataset is directional only):

- **Payment history (Missed_Payments, Month_1–6):** Past missed or late payments are the strongest predictor of future delinquency in credit practice, so these fields must be reconciled before they can be used.
- **Debt-to-income ratio:** It measures how stretched a customer's repayment capacity is; in this data, customers with a ratio of 0.20 or less were delinquent 9.8% of the time vs 17.2% above that level ($p = 0.10$).
- **Credit utilization:** High use of available credit signals financial stress; customers using 30% or less of their limit were delinquent 12.0% of the time vs 16.8% above that level, and the 4 over-limit accounts deserve a separate flag.
- **Account tenure:** New relationships carry more uncertainty; customers with 2 years or less of tenure were delinquent 20.8% of the time vs 15.1% for longer-tenured customers.
- **Employment status:** Income stability affects ability to pay; unemployed customers had the highest delinquency rate (19.4%) and retired customers the lowest (11.5%), although the difference is not significant.

Fairness note: Age (customers aged 46–60 showed 21.7% delinquency) and Location (12.0%–19.6% across cities) should not be used as decision drivers. The age pattern is fragile once many comparisons are considered, and both variables can act as proxies for protected characteristics.

5. AI & GenAI Usage

A GenAI assistant (Claude) was used to profile the dataset, check data quality, diagnose missingness and run the statistical tests behind this report. Every figure was produced by code run on the actual data and then reviewed, rather than taken from unverified AI text. To protect confidentiality, Customer_ID was excluded from analysis outputs and only aggregated statistics are reported.

Example AI prompts used:

- "Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency."
- "For each column with missing values, test whether missingness is related to delinquency or to other variables, and recommend an imputation strategy based on industry best practices."
- "Check for logical inconsistencies between fields, such as missed payments over 12 months versus the six-month payment history, and credit score versus missed payments."
- "Compare delinquency rates across key variables, test whether differences are statistically significant, and identify the top variables most likely to predict delinquency."

6. Conclusion & Next Steps

The dataset is reasonably complete, and its gaps can be handled with median imputation plus missing-value flags. However, inconsistent labels, unit mismatches and contradictions between payment-related fields currently leave the data with almost no measurable predictive signal. Building a model on it as-is would produce unreliable and potentially unfair risk scores. Recommended next steps:

- Confirm with Geldium how Delinquent_Account, Missed_Payments and the monthly history are defined and sourced, and correct the conflicting records.
- Standardize Employment_Status labels, align utilization and debt-to-income units with the data guide, and validate floor values (0.05 utilization, 0.10 debt-to-income).
- Apply the imputation plan in Section 3 and keep the missing-value flags as candidate features.
- In the modeling phase, address the 16% class imbalance (class weighting or resampling within training folds only), evaluate with recall, precision, F1 and AUC rather than accuracy, and run fairness checks on age- and location-related outcomes.